



Automating As-Built and GIS Data Requests

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Overview

- Background
- Request for Automated Tool
- Challenges
 - Other Options and Considerations
 - Permissions Handling
- Solution
 - Survey123
 - Power Automate
 - Python
- Tracking



As-Built Download Request Form

Fill out all required fields and draw a polygon on the map to submit an as-built download request. You will receive an email with the path to the requested as-builts.

Email*

Specify Area of Interest*

Draw a polygon in the map below to submit a GIS as-built request.

Click the polygon tool in the top right of the map to begin.

Esri, NASA, NGA, USGS, FEMA | Esri Community Maps Contributors, County of Dakota, ... Powered by Esri

Perimeter: 0.5 Miles Area: 3.44 Acres

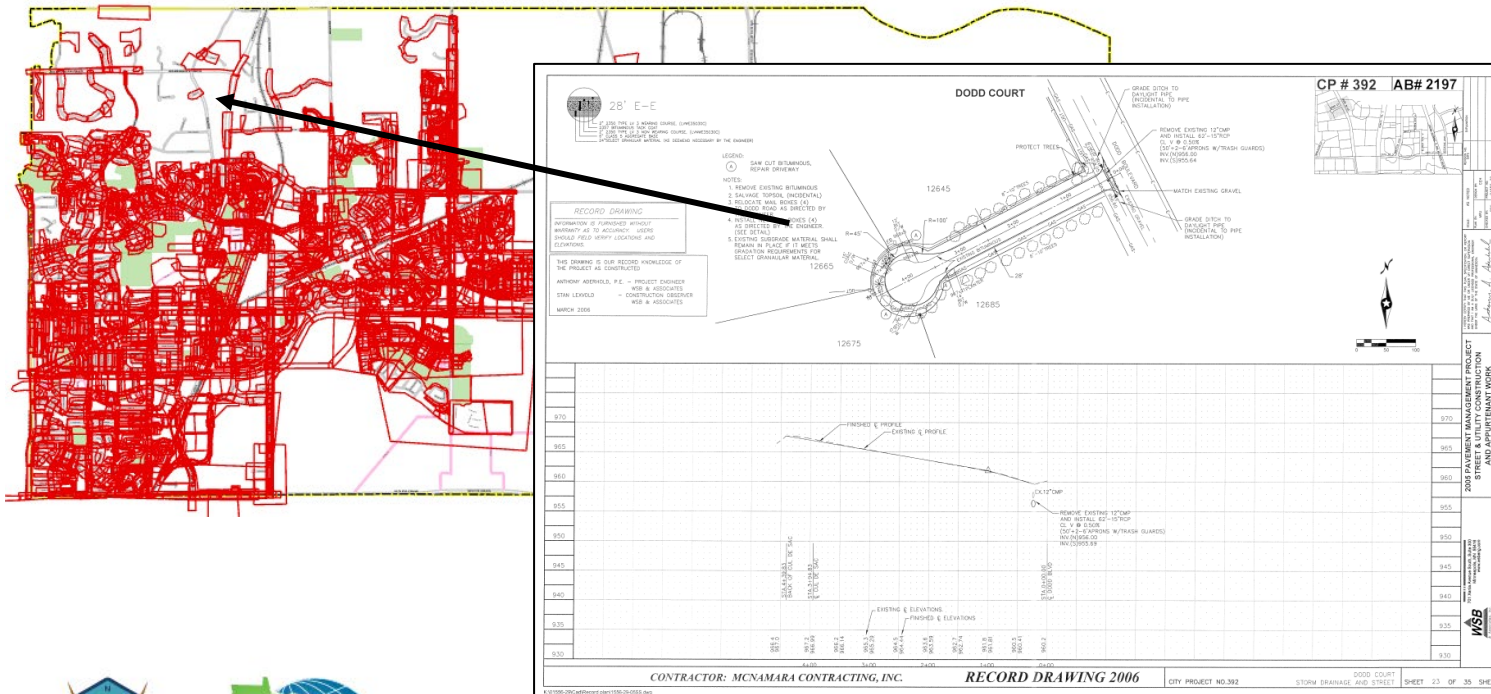
Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Background – Plan area layer

- Utility as-built pages are extracted, stamped, and their extent digitized to a 'Plan Areas' layer



- Each page is saved to a Sharepoint folder with name format:
`{as-built #}.pdf`
- A hyperlink concatenates automatically in the Plan Areas layer using attribute rules

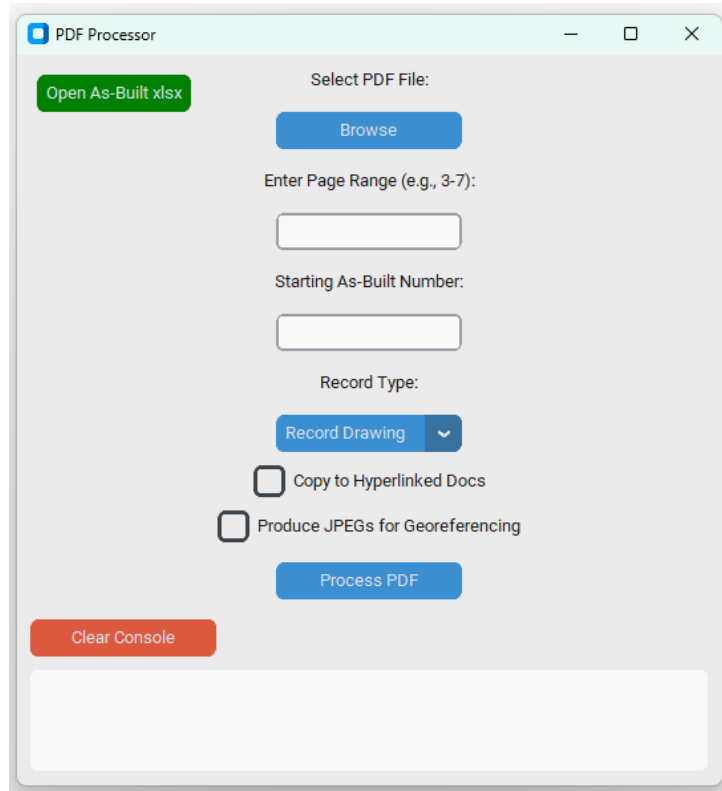


Github: Code/PPT



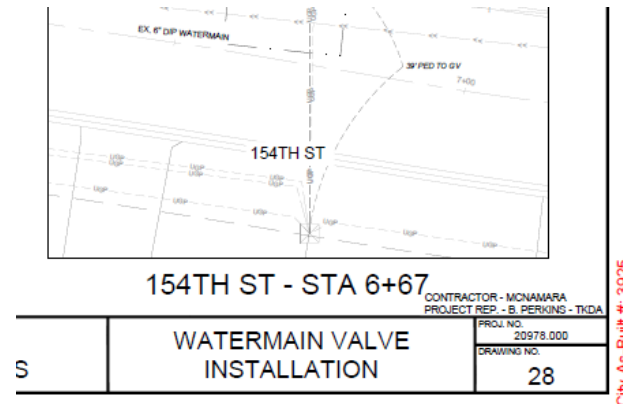
github.com/m-hanson-5/As-Built-Downloader

Plan Area Layer – As-Built Extractor Tool



The screenshot shows a web application titled "PDF Processor". It features a "Select PDF File:" section with a green "Open As-Built.xlsx" button and a blue "Browse" button. Below this is a text input field for "Enter Page Range (e.g., 3-7):". Another text input field is for "Starting As-Built Number:". A "Record Type:" dropdown menu is set to "Record Drawing". There are two checkboxes: "Copy to Hyperlinked Docs" and "Produce JPEGs for Georeferencing". A blue "Process PDF" button is at the bottom. A red "Clear Console" button is located above a large white text area for console output.

1. Stamps specified pages with sequential as-built numbers
2. Extracts each page to pdf
3. Saves pdf's to Sharepoint directory (optional)
4. Produces JPEG's (optional)



https://github.com/m-hanson-5/PDF_Extractor



Python modules: PyPDF2, reportlab, fitz,
PIL, customtkinter

Plan area layer - Fields

- Field Schema:
 - ID (As-Built #; this field is used to relate to other layers' as-built references)
 - Hyperlink
 - Water (y/n)
 - Sanitary (y/n)
 - Storm (y/n)
 - Grading (y/n)
 - Project Number
 - Project Name
 - Install Year
 - Plan Type (Record Drawing, Construction Plan, Private Plan)

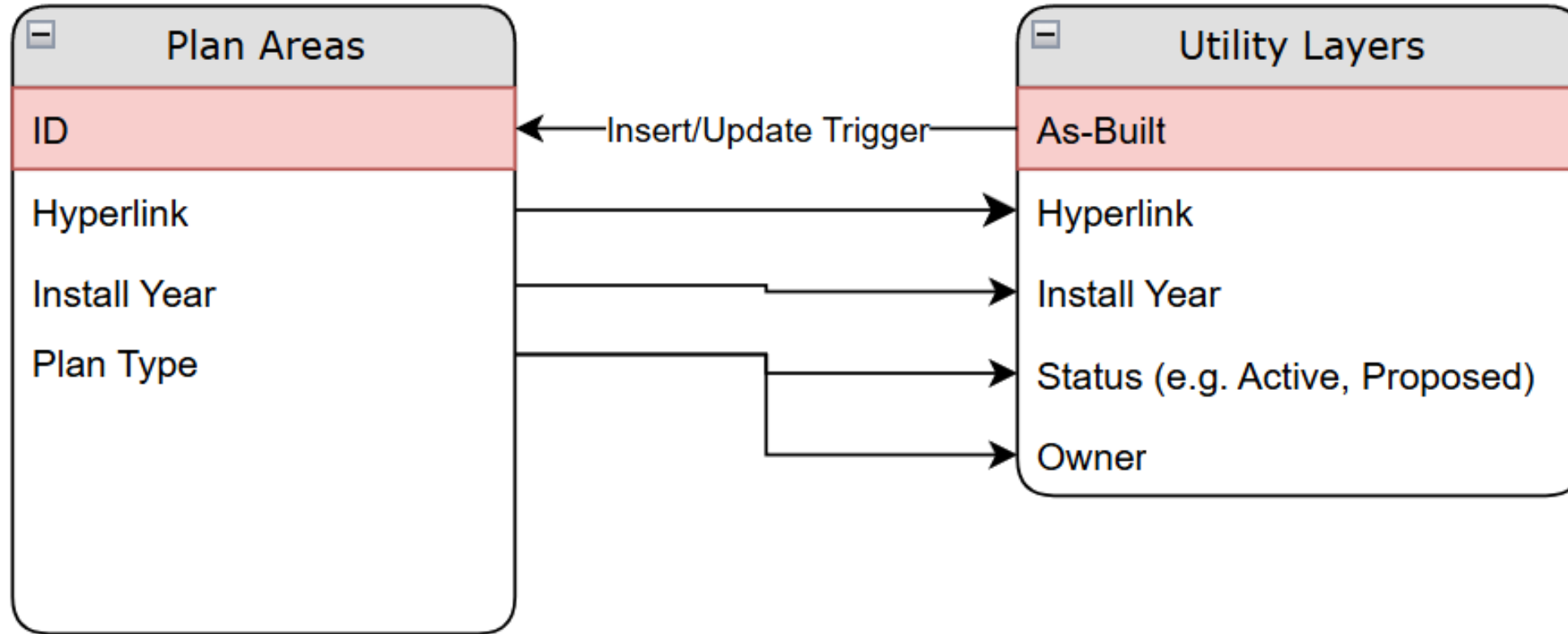


Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Plan area layer – Attribute Rules



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

The Tool

- Requirements:
 - Web interface. No login needed
 - Tool accepts a polygon input
 - Validation process to ensure requester has permission to access data
 - Output is an emailed link to a Sharepoint folder with as-built pdf's and GIS files (if specified)



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Challenges

- Process begins online but must access local files
- Permissions Handling

SOLUTIONS CONSIDERED:

- Add script to a toolbox and share as a Web Tool (Geoprocessing Service)
 - Difficult to customize without Javascript, same issue of file management from web tool
 - Power Automate Premium
 - Allows use of Power Automate Desktop in 'unattended mode'
 - Can run a local script in response to a trigger
 - \$\$\$
- ✓ Python script on a polling interval using Task Scheduler



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

WORKFLOW

Survey123

User draws a polygon
in Survey123

Power Automate

Power Automate detects the submission

- Sends an email to the user
acknowledging submission
- Creates a GIS Ticket for tracking

Python

Python script polls for changes to the
Survey123 layer and processes the
request



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

As-Built Download Request Form

Fill out all required fields and draw a polygon on the map to submit an as-built download request. You will receive an email with the path to the requested as-builts.

Email*

michael.hanson@rosemountmn.gov

Specify Area of Interest*

Draw a polygon in the map below to submit a GIS as-built request.

Click the polygon tool in the top right of the map to begin.

Find address or place

+

-

Home

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Full Screen

Map Tools

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CAMERON WAY

CAMBRIAN AVE

S ROBERT TRL

Esri, NASA, NGA, USGS, FEMA | Esri Community Maps Contributors, County of Dakota, ...

Powered by Esri

① Perimeter: 0.5 Miles Area: 3.44 Acres



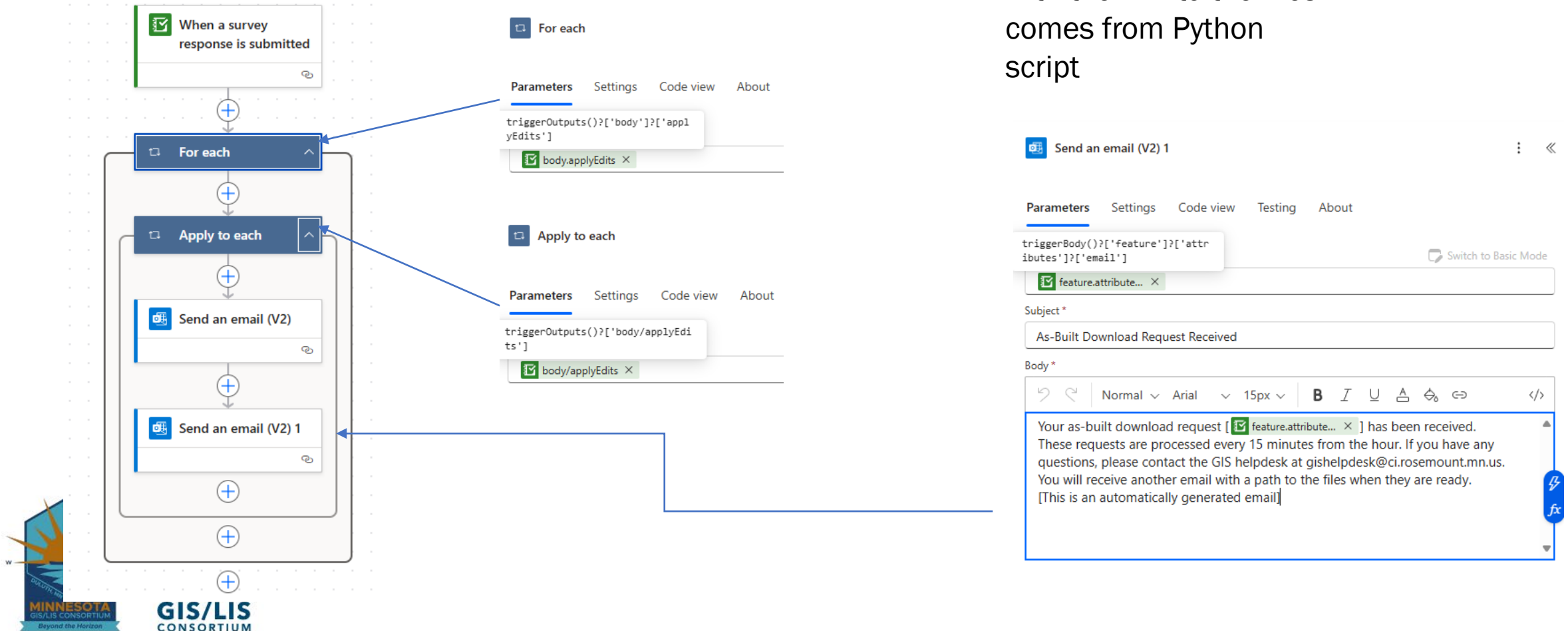
github.com/m-hanson-5/As-Built-Downloader

Power Automate Flow



Github: Code/PPT

Note: the final email with the link to the files comes from Python script



Example of Email to GIS HelpDesk

A new request with request id 200004367 has been created by Public Works. Short info on the request is :

Title : A new as-built download request has been submitted.

Category :

Description :A new as-built download request has been submitted.

Requestor Email:<derick.anderson@rosemountmn.gov>

Desired Output Location: <DCTC>

Global ID: {DD597079-7605-46B6-8CC6-194F0E5BFA98}

You may fulfill the download request immediately by running the below script:

\\Automation\\As-Built_Auto_Downloader.py

[As-Built Request Tracker](#)

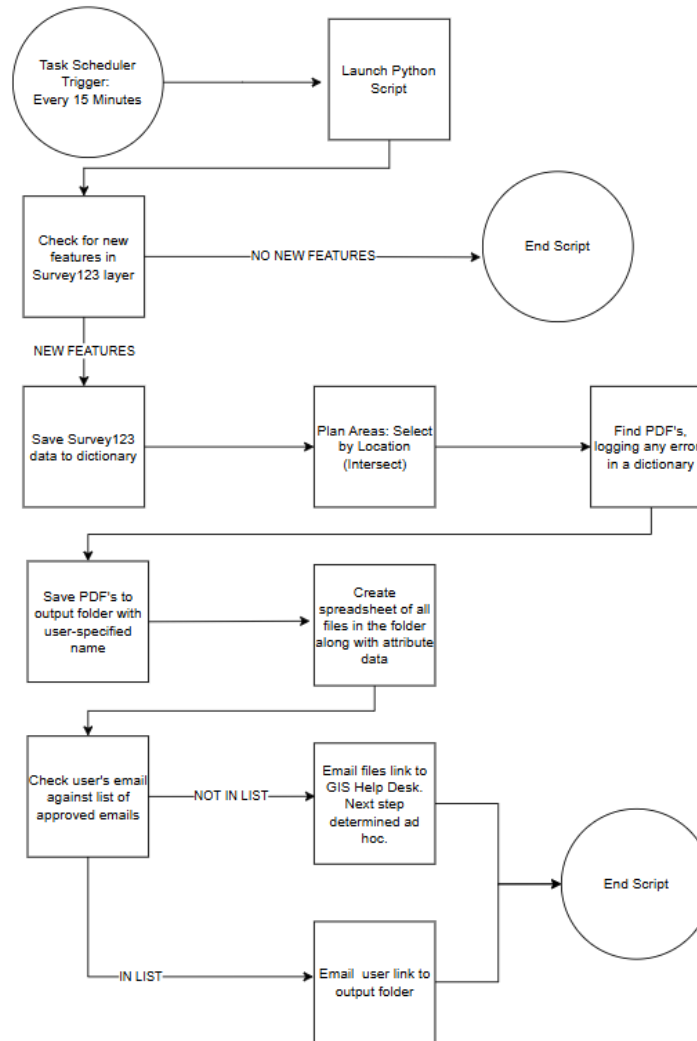


Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Python



View Code on GitHub:

<https://github.com/m-hanson-5/As-Built-Downloader>



Python modules: arcpy, pandas, logging

Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Python

AS-BUILT DOWNLOADER

> logs

✓ src

> __pycache__

models.py

process_as_built.py

process_gis_files.py

config.json

launch_main.vbs

main.py

README.md

setup.py

```
fields = ["GlobalId", "FulfilledDate", "utilities",
          "email", "specify_desired_output_folder_n", "desired_output", "CreationDate"]

## unpacking from esri into SurveyRecord dataclass, stored in dict with globalid as key
surveyData = {}
with arcpy.da.SearchCursor(survey, fields) as cursor:
    for row in cursor:
        if row[1] is None: # FulfilledDate is None (i.e. not fulfilled)

            # Create a dictionary from the row values and field names
            record_dict = dict(zip(fields, row))

            # Create a SurveyRecord dataclass instance from the dict using argument unpacking
            survey_record = SurveyRecord(**record_dict)
            globalid = survey_record.GlobalId.strip() # Clean up the GlobalId

            # Store it in a dictionary keyed by globalid
            surveyData[globalid] = survey_record

# if no valid globalIDs are found, notify the user and exit
if not surveyData:
    print("No valid GlobalIDs found after checking the FulfilledDate field. Exiting Script.")
    logging.info("No valid GlobalIDs found after checking the FulfilledDate field. Exiting Script.")
    logging.shutdown()
    time.sleep(1) # Short delay before deleting the log file

    # Delete the log file
    if os.path.exists(log_file):
        os.remove(log_file)

sys.exit()
```



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Python

```
@dataclass
class SurveyRecord:
    GlobalId: str
    email: str
    desired_output: str
    CreationDate: Optional[datetime] = None
    FulfilledDate: Optional[datetime] = None
    as_built_fulfilled: Optional[datetime] = None
    gis_files_fulfilled: Optional[datetime] = None
    specify_desired_output_folder_n: str = ""
    folder: str = "" # the above field will store here
    utilities: str = ""
    utilities_other: str = ""
    other: str = ""
    other: bool = False
    as_built: bool = False
    gis_files: bool = False
    utilities_list: list = field(default_factory=list)
    outputs_list: list = field(default_factory=list)
```

```
def __post_init__(self):
    self.GlobalId = self.GlobalId.strip()

    # Set the folder name based on long field name
    if self.specify_desired_output_folder_n:
        self.folder = self.specify_desired_output_folder_n.strip()

    # Also, set as_built and gis_files flags from desired_output
    self.outputs_list = self.desired_output.lower().split(",") if self.desired_output else []
    self.as_built = 'as_built' in self.outputs_list
    self.gis_files = 'gis_files' in self.outputs_list
    self.utilities_list = self.utilities.split(",") if self.utilities else []
```



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Python – Managing Fields Shared in GIS Files

```
# Get layers to process and their fields
layer_dict = {'Curb Stop': ("FacilityID", "Diameter", "InstallYear", "LifeCycleStatus"),
              'System Valve': ("FacilityID", "Diameter", "ValveType", "WaterType", "LifeCycleStatus"),
              'Fire Hydrant': ("FacilityID", "LifeCycleStatus", "WaterType"),
              'Water Main': ("FacilityID", "LifeCycleStatus", "Material", "Diameter", "WaterType"),
              'Water Lateral': ("FacilityID", "LifeCycleStatus", "Material", "Size", "Type", "WaterType"),
              'Sanitary Clean Out': ("FacilityID", "LifeCycleStatus", "TopElev"),
              'Sanitary Gravity Main': ("FacilityID", "LifeCycleStatus", "Material", "Diameter", "InstallYear"),
              'Sanitary Force Main': ("FacilityID", "LifeCycleStatus", "Material", "Size", "InstallYear"),
              'Sanitary Lateral Line': ("FacilityID", "Material", "Diameter", "InstallYear", "LifeCycleStatus"),
              'Sanitary Manhole': ("FacilityID", "LifeCycleStatus", "InstallYear", "TopElev"),
              'Sanitary Valve': ("FacilityID", "LifeCycleStatus"),
              'Sanitary Lift Station': ("FacilityID"),
              'Storm Mains': ("FacilityID", "Material", "Measurement1", "InstallDate", "LifeCycleStatus"),
              'Storm Inlets': ("FacilityID", "LifeCycleStatus", "Type", "InstallDate"),
              'Storm Forcemains': ("FacilityID", "Material", "Measurement1", "InstallDate", "LifeCycleStatus"),
              'Storm Lift Station': ("FacilityID", "LifeCycleStatus"),
              'Storm Manholes': ("FacilityID", "LifeCycleStatus", "Type"),
              'Storm Outlets': ("FacilityID", "Type", "Sump", "LifeCycleStatus")}

water_layers = ["Curb Stop", "System Valve", "Fire Hydrant", "Water Main", "Water Lateral"]
sanitary_layers = ["Sanitary Clean Out", "Sanitary Gravity Main", "Sanitary Force Main", "Sanitary Lateral Line",
                  "Sanitary Manhole", "Sanitary Valve", "Sanitary Lift Station"]
storm_layers = ["Storm Mains", "Storm Inlets", "Storm Forcemains", "Storm Lift Station", "Storm Manholes", "Storm Outlets"]
```

```
layers_list = []
if "All" in Record.utilities_list or Record.utilities_list is None:
    layers_list.extend(layers)
else:
    if "Water" in Record.utilities_list:
        layers_list.extend(water_layers)
    if "Sanitary" in Record.utilities_list:
        layers_list.extend(sanitary_layers)
    if "Storm" in Record.utilities_list:
        layers_list.extend(storm_layers)
```



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader

Task Scheduler – Silent Operation

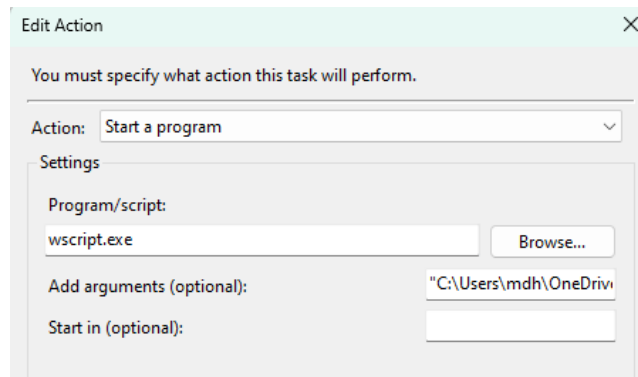
- Method with a .vbs file below, though .bat is also possible

vbs file content:

```
Set WshShell = WScript.CreateObject("WScript.Shell")
WshShell.Run """C:\Program Files\ArcGIS\Pro\bin\Python\envs\arcgispro-py3\python.exe""" """C:\Users\mdh\OneDrive - ci.rosemount.mn.us\GIS Projects\Automations\As-Built_Auto_Downloader.py""", 0
```

.vbs file

Task Scheduler:



Path to .vbs file

Alternatives for silent operation:

- 'Run whether user is logged in or not'
- Pythonw.exe
- 'start /B' in .bat file

Github: Code/PPT



Tracker App

As-Built Download Request Tracker

- Experience Builder app to track open tickets

GH GIS Helpdesk
To: Michael Hanson; Aaron Menza

A new request with request id 200004809 has been created by Public Works. Short info on the request is :

Title : A new as-built download request has been submitted.
Category :
Description :
A new as-built download request has been submitted.

Requestor Email:<derick.anderson@rosemountmn.gov>

Desired Output Location: <e.g. 20240726_BonairePathExtension>

Global ID: {F5256FC8-450E-405B-80EE-6AA1A67C0007}

You may fulfill the download request immediately by running the below script:

Automation\AS-Built_Auto_Downloader.py

[As-Built Request Tracker](#)

[illegible]

Questions?



Github: Code/PPT



github.com/m-hanson-5/As-Built-Downloader